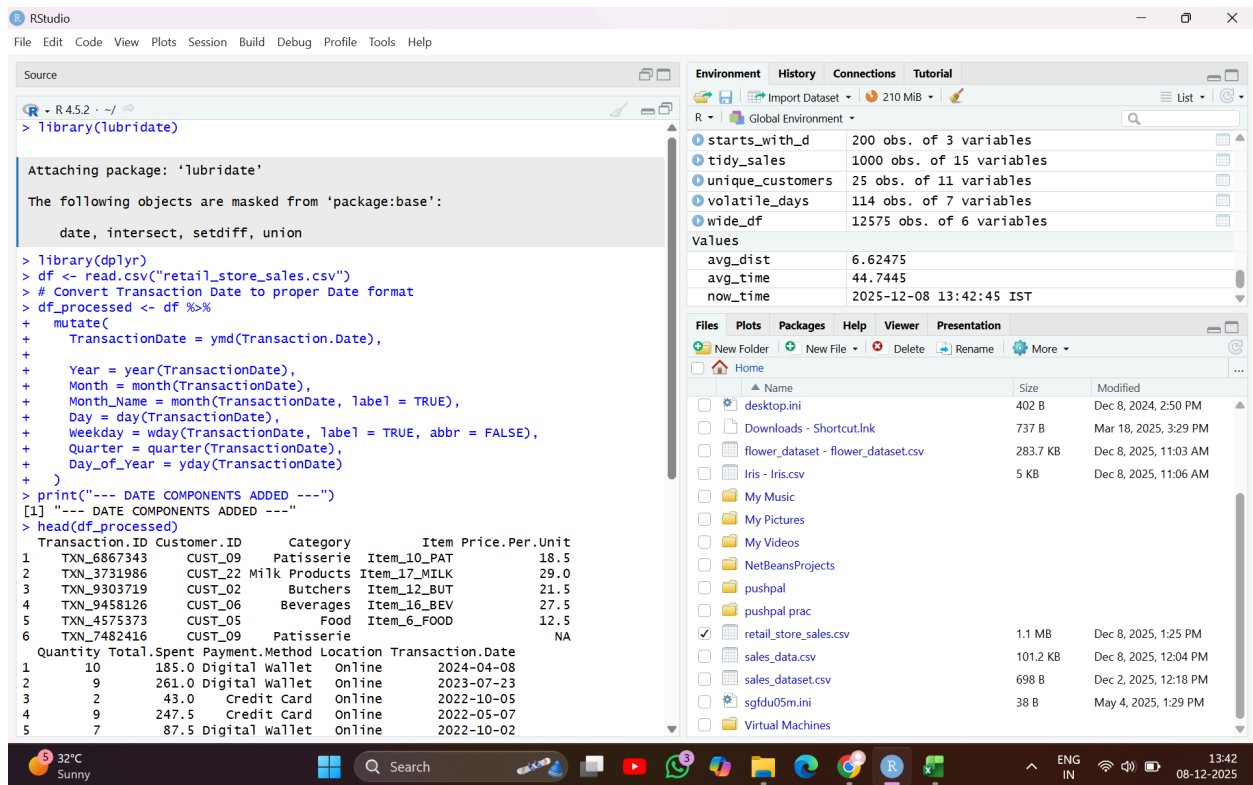


SHETH L.U.J AND SIR M.VCOLLEGE

Subject: Data Analysis with SAS / SPSS /R

Practical no. 14

Aim: Extracting date components
using lubridate:: functions (R).



The screenshot shows the RStudio interface with the following components:

- Source:** R console output showing the loading of the lubridate package and the execution of code to read a CSV file, convert the TransactionDate to a proper Date format, and extract date components (Year, Month, Day, Weekday, Quarter, Day_of_Year).
- Environment:** A list of objects in the Global Environment, including starts_with_d, tidy_sales, unique_customers, volatile_days, and wide_df.
- Files:** A file explorer showing the contents of the Home directory, including files like desktop.ini, Downloads - Shortcut.Lnk, flower_dataset - flower_dataset.csv, Iris - Iris.csv, My Music, My Pictures, My Videos, NetBeansProjects, pushpal, pushpal prac, retail_store_sales.csv, sales_data.csv, sales_dataset.csv, sgfd05m.ini, and Virtual Machines.

The console output shows the following code and results:

```
> library(lubridate)

Attaching package: 'lubridate'

The following objects are masked from 'package:base':

    date, intersect, setdiff, union

> library(dplyr)
> df <- read.csv("retail_store_sales.csv")
> # Convert Transaction Date to proper Date format
> df_processed <- df %>%
+   mutate(
+     TransactionDate = ymd(Transaction.Date),
+     Year = year(TransactionDate),
+     Month = month(TransactionDate),
+     Month_Name = month(TransactionDate, label = TRUE),
+     Day = day(TransactionDate),
+     Weekday = wday(TransactionDate, label = TRUE, abbr = FALSE),
+     Quarter = quarter(TransactionDate),
+     Day_of_Year = yday(TransactionDate)
+   )
> print("--- DATE COMPONENTS ADDED ---")
[1] "--- DATE COMPONENTS ADDED ---"
> head(df_processed)
  Transaction.ID Customer.ID Category Item Price.Per.Unit
1 TXN_6867343 CUST_09 Patisserie Item_10_PAT 18.5
2 TXN_3731986 CUST_22 Milk Products Item_17_MILK 29.0
3 TXN_9303719 CUST_02 Butchers Item_12_BUT 21.5
4 TXN_9458126 CUST_06 Beverages Item_16_BEV 27.5
5 TXN_4575373 CUST_05 Food Item_6_FOOD 12.5
6 TXN_7482416 CUST_09 Patisserie NA NA

  Quantity Total.Spent Payment.Method Location Transaction.Date
1 10 185.0 Digital wallet Online 2024-04-08
2 9 261.0 Digital wallet Online 2023-07-23
3 2 43.0 Credit Card Online 2022-10-05
4 9 247.5 Credit Card Online 2022-05-07
5 7 87.5 Digital wallet Online 2022-10-02
```

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Source

```
R - R4.5.2 - ~/
> print("--- DATE COMPONENTS ADDED ---")
[1] "--- DATE COMPONENTS ADDED ---"
> head(df_processed)
  Transaction.ID Customer.ID Category Item Price.Per.Unit
1 TXN_6867343 CUST_09 Patisserie Item_10_PAT 18.5
2 TXN_3731986 CUST_22 Milk Products Item_17_MILK 29.0
3 TXN_9303719 CUST_02 Butchers Item_12_BUT 21.5
4 TXN_9458126 CUST_06 Beverages Item_16_BEV 27.5
5 TXN_4575373 CUST_05 Food Item_6_FOOD 12.5
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  Quantity Total.Spent Payment.Method Location Transaction.Date
1 10 185.0 Digital wallet online 2024-04-08
2 9 261.0 Digital wallet online 2023-07-23
3 2 43.0 Credit Card online 2022-10-05
4 9 247.5 Credit Card online 2022-05-07
5 7 87.5 Digital wallet online 2022-10-02
6 10 200.0 Credit Card online 2023-11-30
  Discount.Applied TransactionDate Year Month Month_Name Day Weekday
1 True 2024-04-08 2024 4 Apr 8 Monday
2 True 2023-07-23 2023 7 Jul 23 Sunday
3 False 2022-10-05 2022 10 Oct 5 Wednesday
4 False 2022-05-07 2022 5 May 7 Saturday
5 False 2022-10-02 2022 10 Oct 2 Sunday
6 False 2023-11-30 2023 11 Nov 30 Thursday
  Quarter Day_of_Year
1 2 99
2 3 204
3 4 278
4 2 127
5 4 275
6 4 334
> # CURRENT SYSTEM DATE
> now_time <- now()
> print(paste("Current Year:", year(now_time)))
[1] "Current Year: 2025"
> print(paste("Current Month:", month(now_time)))
[1] "Current Month: 12"
> print(paste("Current Hour:", hour(now_time)))
[1] "Current Hour: 13"
>
```

Environment History Connections Tutorial

R - Global Environment

- starts_with_d 200 obs. of 3 variables
- tidy_sales 1000 obs. of 15 variables
- unique_customers 25 obs. of 11 variables
- volatile_days 114 obs. of 7 variables
- wide_df 12575 obs. of 6 variables

Values

- avg_dist 6.62475
- avg_time 44.7445
- now_time 2025-12-08 13:42:45 IST

Files Plots Packages Help Viewer Presentation

New Folder New File Delete Rename More

Home

Name	Size	Modified
desktop.ini	402 B	Dec 8, 2024, 2:50 PM
Downloads - Shortcut.lnk	737 B	Mar 18, 2025, 3:29 PM
flower_dataset - flower_dataset.csv	283.7 KB	Dec 8, 2025, 11:03 AM
Iris - Iris.csv	5 KB	Dec 8, 2025, 11:06 AM
My Music		
My Pictures		
My Videos		
NetBeansProjects		
pushpal		
pushpal prac		
retail_store_sales.csv	1.1 MB	Dec 8, 2025, 1:25 PM
sales_data.csv	101.2 KB	Dec 8, 2025, 12:04 PM
sales_dataset.csv	698 B	Dec 2, 2025, 12:18 PM
sgfdu05m.ini	38 B	May 4, 2025, 1:29 PM
Virtual Machines		

32°C Sunny

Search

ENG IN

13:43 08-12-2025